(v) If $\theta = \frac{\pi}{2}$ then $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\hat{n}$.

(vi) For any scalars m and n,

$$m\vec{a} \times n\vec{b} = mn(\vec{a} \times \vec{b}) = (mn\vec{a}) \times \vec{b} = \vec{a} \times (mn\vec{b}) = n\vec{a} \times m\vec{b} \ .$$

(vii) Vector product is distributive over addition.

That is, $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$

$$(\vec{a} + \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) + (\vec{b} \times \vec{c}).$$

This property can be extended to subtraction and to any number of vectors.

That is, $\vec{a} \times (\vec{b} - \vec{c}) = (\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c})$

$$\vec{a} \times (\vec{b} + \vec{c} + \vec{d}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}) + (\vec{a} \times \vec{d}).$$

(viii) Working rule to find the cross product

$$\text{Let} \quad \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} \ , \quad \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k} \ .$$

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \\ &= a_1b_1(\hat{i} \times \hat{i}) + a_1b_2(\hat{i} \times \hat{j}) + a_1b_3(\hat{i} \times \hat{k}) + a_2b_1(\hat{j} \times \hat{i}) \\ &\quad + a_2b_2(\hat{j} \times \hat{j}) + a_2b_3(\hat{j} \times \hat{k}) + a_3b_1(\hat{k} \times \hat{i}) + a_3b_2(\hat{k} \times \hat{j}) + a_3b_3(\hat{k} \times \hat{k}) \\ &= (a_2b_3 - a_3b_2)\hat{i} - (a_1b_3 - a_3b_1)\hat{j} + (a_1b_2 - a_2b_1)\hat{k} \end{aligned}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} .$$

(ix) If θ is the angle between $\vec{a}$ and $\vec{b}$ then $\theta = \sin^{-1}\left[\frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}\right]$.

(The proof of this result is left as an exercise)

Note 8.7

In this case θ is always acute. Thus, if we try to find the angle using vector product, we get only the acute angle. Hence in problems of finding the angle, the use of dot product is preferable since it specifies the position of the angle θ.

(x) The unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ are $\pm\hat{n} = \pm\frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$ (how?)

Vectors of magnitude λ, perpendicular to both $\vec{a}$ and $\vec{b}$ are $\pm\lambda\hat{n} = \pm\lambda\frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$.

Example 8.20

Find $|\vec{a} \times \vec{b}|$, where $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = \hat{i} + \hat{j} + \hat{k}$.

Solution

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(4-0) - \hat{j}(3-0) + \hat{k}(3-4) = 4\hat{i} - 3\hat{j} - \hat{k} \ .$$

$$|\vec{a} \times \vec{b}| = |4\vec{i} - 3\vec{j} - \vec{k}| = \sqrt{16+9+1} = \sqrt{26} \ .$$

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